

Lecture 4 p.1

$$C \frac{dT}{dt} = F_{in} - F_{out}$$

$$\epsilon \in [0, 1]$$

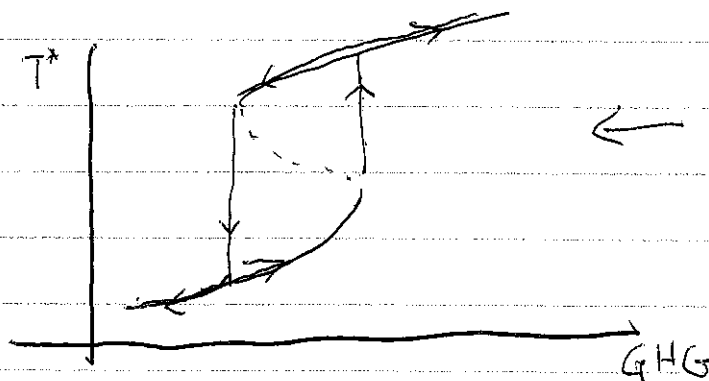
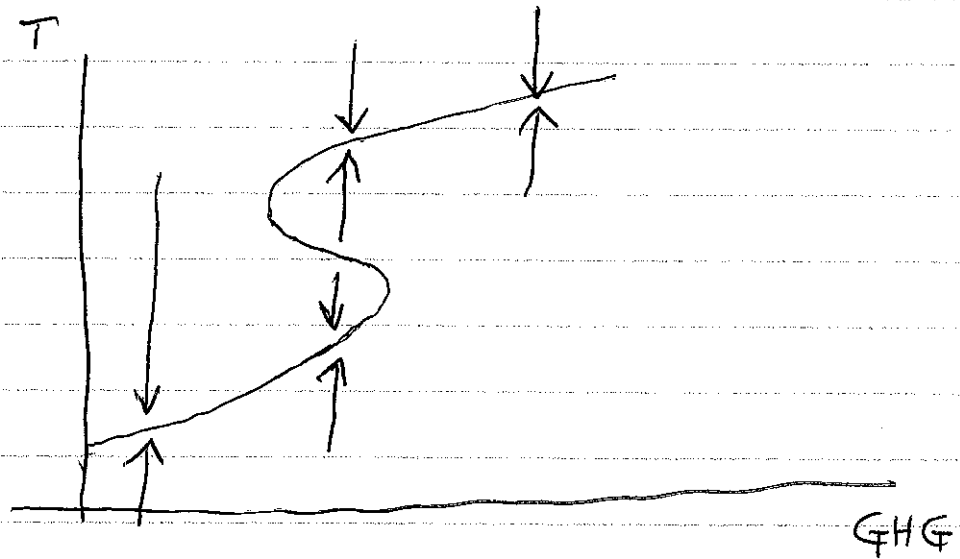
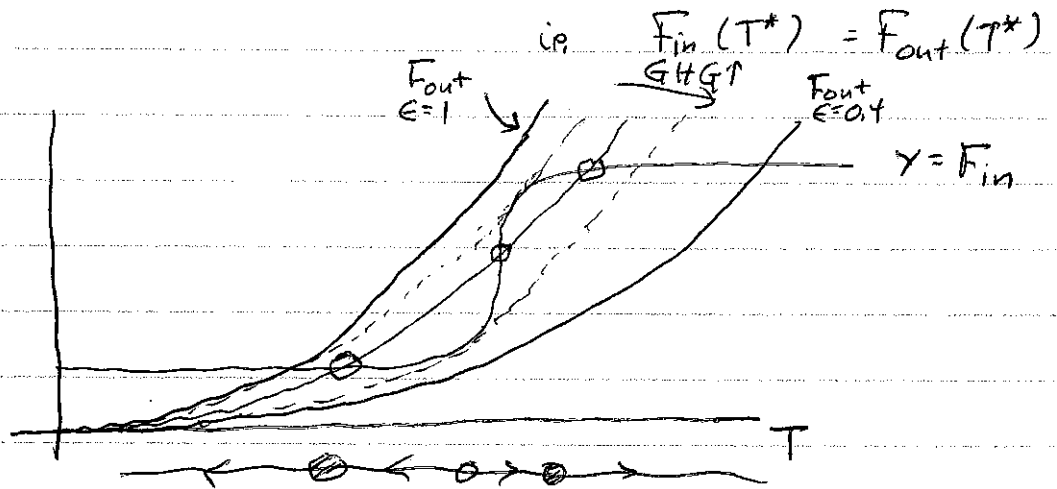
$$F_{in} = (1 - \alpha(T)) Q$$

$$F_{out} = \epsilon \sigma T^4$$

equilibria T^* satisfy

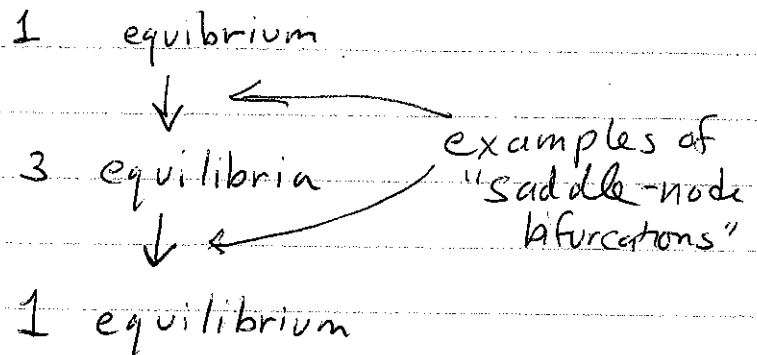
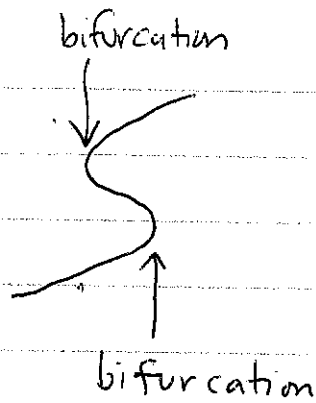
$$F_{in}(T^*) - F_{out}(T^*) = 0$$

$$F_{in}(T^*) = F_{out}(T^*)$$



← Bifurcation Diagram demonstrating hysteresis loop

Lecture 4 p.2



Bifurcations in 1-D:

$$\ddot{x} = f(x; \lambda)$$

$x \in \mathbb{R}$, f smooth enough,
 $\lambda \in \mathbb{R}$ is a "bifurcation parameter"

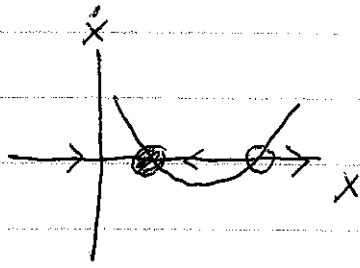
Only qualitative options for solns:

- equilibrium — stay put
- diverge to $\pm \infty$
- approach, monotonically, an equilibrium

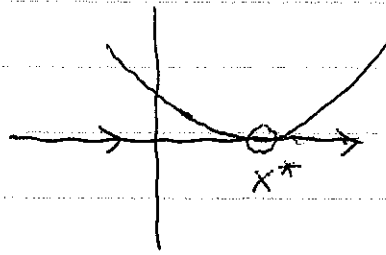
Possible changes with parameter:

- lose an equilibrium or gain one
- equilibrium changes its stability property —
e.g. stable to unstable

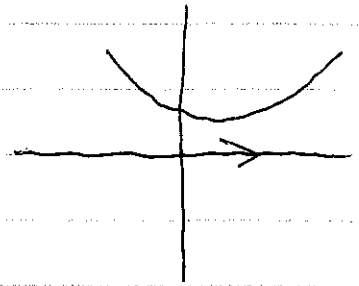
example



$\lambda < \lambda_c$
2 equilibria



$\lambda = \lambda_c$
 $f(x^*; \lambda_c) = 0$
 $f'(x^*; \lambda_c) = 0$



$\lambda > \lambda_c$
0 equilibria

Implicit Function Thm.

Let $f: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ be C^1 with coordinates (x, λ) , i.e. $x \in \mathbb{R}^n, \lambda \in \mathbb{R}^m$. Fix a pt. (x^*, λ^*) with $f(x^*, \lambda^*) = 0$. If Jacobian matrix $J(x^*, \lambda^*) = \left[\frac{\partial f_i}{\partial x_j}(x^*, \lambda^*) \right]$ is invertible, then there exists an open neighborhood U of $\mathbb{R}^m$ containing λ^* s.t. there exists a unique C^1 function $g: U \rightarrow \mathbb{R}^n$ with $g(\lambda^*) = x^*$ and $f(x(\lambda), \lambda) = 0$ for all $\lambda \in U$.

Note: J is invertible if $\text{Det}(J) \neq 0$.

example in 1-d

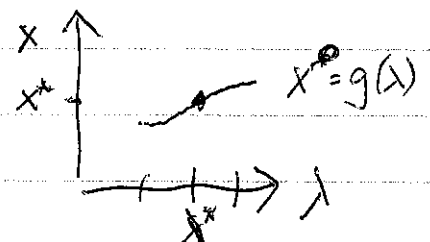
$$\dot{x} = f(x, \lambda), \quad x, \lambda \in \mathbb{R} \quad (n=m=1)$$

$$f(x^*, \lambda^*) = 0$$

$$f'(x^*, \lambda^*) \neq 0$$

$$\Rightarrow \exists \text{ unique } g(\lambda) = x, \lambda \in U,$$

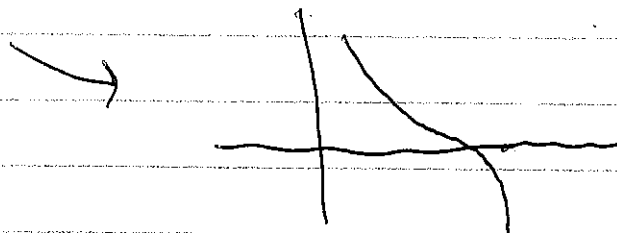
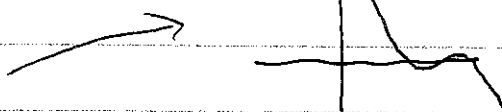
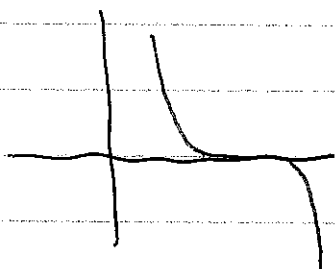
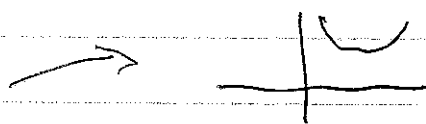
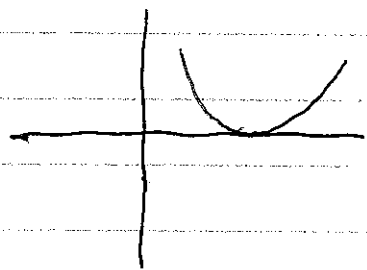
$$x^* = g(\lambda^*), \quad f(g(\lambda), \lambda) = 0$$



Bifurcations in 1D occur at (x^*, λ^*) if

$$\left. \begin{aligned} f(x^*, \lambda^*) &= 0 \\ \frac{df}{dx}(x^*, \lambda^*) &= 0 \end{aligned} \right\}$$

- linear stability inconclusive
- implicit function thm. doesn't apply
- system is sensitive to qualitative changes with ~~varies~~ small changes to parameters (not robust)



Taylor-expansion of $f(x, \lambda)$ about $x = x^*$ & $\lambda = \lambda^*$:

$$f(x, \lambda) = f(x^* + y, \lambda^* + \mu)$$

$$|y|, |\mu| \ll 1$$

Lecture 4 p.5

$$f(x^*+y, \lambda^*+\mu)$$

$$= \cancel{f(x^*, \lambda^*)}^0 + \cancel{\frac{\partial f}{\partial x}(x^*, \lambda^*)}^0 y$$

$$+ \underbrace{\frac{\partial f}{\partial \lambda}(x^*, \lambda^*)}_{\alpha} \mu + \underbrace{\frac{1}{2} \frac{\partial^2 f}{\partial x^2}(x^*, \lambda^*)}_{\beta} y^2 + \dots$$

Leading order behavior in (y, μ) if $\alpha, \beta \neq 0$

$$\frac{dy}{dt} = \alpha \mu + \beta y^2$$

$$\text{let } t = a\tau, y = bY, \quad a > 0$$

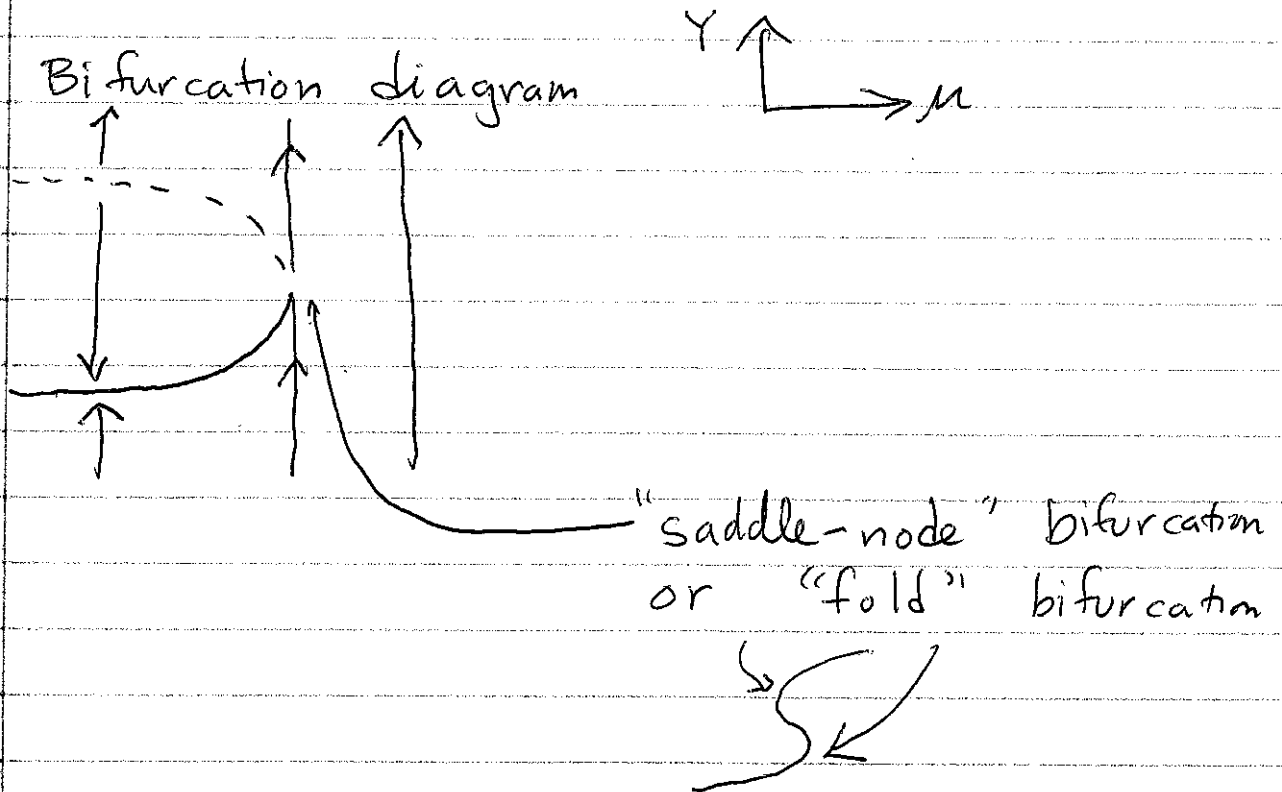
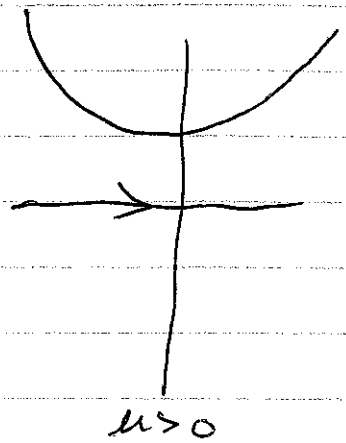
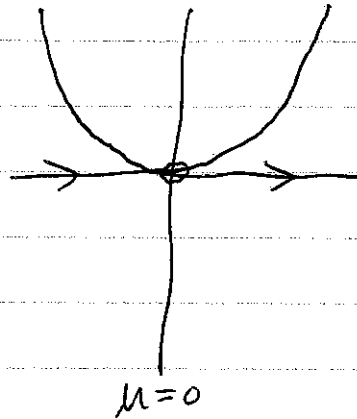
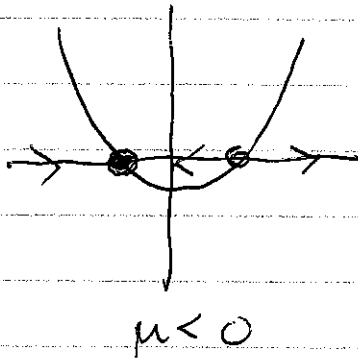
$$\frac{dy}{dt} = \frac{d(bY)}{d\tau} \frac{d\tau}{dt} = \frac{b}{a} \frac{dY}{d\tau} = \alpha \mu + \beta b^2 Y^2$$

$$\frac{dY}{d\tau} = \frac{\alpha a}{b} \mu + \underbrace{\beta a b}_{=+1} Y^2$$

$$\Rightarrow \frac{\alpha a}{b} = \frac{\alpha a}{b} (\beta a b) = \alpha \beta a^2 = \pm 1$$

$$\frac{dY}{d\tau} = \pm \mu + Y^2$$

$$\frac{dY}{d\tau} = \mu + Y^2$$



defining conditions:

$$f(x^*, \lambda^*) = \frac{\partial f}{\partial x}(x^*, \lambda^*) = 0$$

&

$$\frac{\partial f}{\partial \lambda}(x^*, \lambda^*) \neq 0, \quad \frac{\partial^2 f}{\partial x^2}(x^*, \lambda^*) \neq 0$$